MILITARY SPECIFICATION

INTERCOM, HELICOPTER AIRCREWMAN, CHEMICAL, BIOLOGICAL, RADIOLOGICAL PROTECTIVE SET

This specification is approved for use within the Naval Air System Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification establishes the requirements for the manufacture and acceptance of a small, light weight, battery powered, intercom to be used with and form part of the Protective Assembly, Helicopter Aircrewman, Chemical, Biological, Radiological (PASS).

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications standards, and handbooks, form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation.

SPECIFICATIONS

FEDERAL

PPP-B-636	-	Box, Shipping, Fiberboard.
PPP-C-795	-	Cushioning Material, Flexible Cellular Plastic Film for Packaging Applications.
PPP-C-1842	-	Cushioning Material, Plastic, Open Cell (For Packaging Applications).

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commanding Officer, Naval Air Engineering Center, Systems Engineering and Standardization Department (Code 53), Lakehurst, NJ 08733-5100, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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MIL-P-116	-	Preservation, Methods of.
MIL-B-117	-	Bags, Sleeves and Tubing-Interior Packaging
MIL-E-5400	-	General Specification for Aerospace Electronic Equipment.
MIL-A-8625	-	Anodic Coatings for Aluminum and Aluminum Alloys.
MIL-C-9177/1	-	Connector, Audio, Airborne, Plug, 4 Contact.
MIL-W-16878	-	General Specification for Insulated Electrical Wire.
MIL-N-18307	-	Nomenclature and Indentification for Aeronautical Systems Including Joint Electronics Type Designated Systems and Associated Support Systems.
MIL-P-28809	-	Printed Wiring Assemblies.
MIL-P-55110	-	Printed Wiring Boards.
MIL-P-85833	-	Protective Assembly, Helicopter Aircrewman, Chemical, Biological, Radiological.

STANDARDS

FEDERAL

FED-STD-595	-	Colors.

MILITARY

DOD-STD-100	-	Engineering Drawing Practices.
MIL-STD-105	-	Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	-	Marking for Shipment and Storage
MIL-STD-130	-	Identification Marking of U.S. Military Property.
MIL-STD-143	-	Order of Precedence for the Selection of Standards and Specifications.
MIL-STD-210	-	Climatic Extremes for Military Equipment.
MIL-STD-454	-	Standard General Requirements for Electronic Equipment.

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DOD-STD-480	-	Configuration Control - Engineering Changes, Deviations and Waivers.
MIL-STD-483	-	Configuration Management Practices for Systems, Equipment, Munitions and Computer Programs.
MIL-STD-490	-	Specification Practices.
MIL-STD-781	-	Reliability Design Qualification and Production Acceptance Tests: Exponential Distribution.
MIL-STD-794	-	Parts and Equipment, Procedures for Packaging of.
MIL-STD-810	-	Environmental Test Methods and Engineering Guidelines.
MIL-STD-889	-	Dissimilar Metals.
MIL-STD-1189	-	Standard Department of Defense Bar Code Symbology

HANDBOOKS

MIL-HDBK-695	-	Rubber Products: Recommended Shelf Life.

2 1.2 Other Government documents The following other Government documents forms a part of this specification to the extent specified herein. Unless otherwise specified, the issues shall be those in effect on the date of the solicitation.

Naval Air Systems Command

SD-24	-	General Specification for Design and Construction of Aircraft Weapon Systems.
NAVAIR 13-1-6 7	-	Aviation-Crew Systems Aircrew Personnel Protective Equipment.

Headquarters, Department of the Army

Army Field Manual FM 3-5	-	NBC Decontamination.

(Copies of specifications, standards, handbooks and other Government documents required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

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2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted shall be those listed in the issue of the DODISS specified in the solicitation. Unless otherwised specified, the issues of documents not listed in the DODISS shall be the issue of the nongovernment documents which is current on the date of the solicitation.

AEROSPACE MATERIALS SPECIFICATIONS

AMS-3238	-	Butyl Rubber, Phosphate Ester Resistant.

(Application for copies should be addressed to the Society of Automotive Engineers, 400 Commonwealth Drive, Warrendale, PA 15096.)

AMERICAN SOCIETY FOR TESTING AND MATERIALS

ASTM D 3951	-	Commercial Packaging.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

2 3 Order of precedence In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained

3. REQUIREMENTS

3 1 Qualification Intercoms furnished under this specification shall be products which are authorized by the qualifying activity for listing on the applicable qualified products list at the time set for the opening of bids (see 4.3 and 6 3).

3.2 First article. When specified in the contract or purchase order, a sample shall be subjected to first article inspection (see 4.4 and 6.4).

3.3 Selection of specifications and standards. Specifications and standards for necessary commodities and services not specified herein shall be selected as specified in MIL-STD-143.

3.4 Materials Materials referenced herein shall conform to applicable specifications and standards. Materials that are not covered by specifications and that are not specifically described herein shall be the best quality and the lightest practical weight. Materials shall be corrosion resistant, or suitably treated to resist corrosion due to fuels, salt spray, environmental, chemical or biological warfare conditions.

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3.4.1 Aluminum alloy. Aluminum alloy parts shall be anodically treated as specified in MIL-A-8625. To provide adequate bonding, the aluminum oxide film formed by this treatment shall be removed from the actual contact area of all surfaces that act as a path for electrical current and from the areas under fasteners used for assembly or mounting purposes.

3.4.2 Dissimilar metals. Dissimilar metals shall not be used in intimate contact with each other. Dissimilar metals are as specified in MIL-STD-889.

3.4.3 Nonmetalic materials. Rubber seals and o-rings shall meet the requirements as specified in AMS-3238 and the requirements specified herein.

3.4.4 Fungus-proof materials. All materials shall be incapable of supporting fungoid growth.

3.4.5 Age. Elastomeric components shall be comprised of materials with expected shelf life of greater than 5 years as specified in MIL-HDBK-695 when consistent with the properties required for functional suitability. These components alone or as part of an assembly shall be delivered to the Government with less than 18 months elapsed from date of cure/manufacture to the date of delivery to the Government.

3.4.6 Toxic materials. The use of materials which produce toxic effects under any conditions shall be avoided.

3.4.7 Flammable materials. The use of materials which will support combustion shall be avoided.

3.5 Design. The intercom shall be used with and form part of the protective assembly (PASS), as shown in Figure 1. The design and construction of the intercom shall be as specified in MIL-E-5400 and the specifications listed herein The intercom shall meet the requirements as specified in MIL-P-85833 and this specification The intercom shall be designed as shown on Figure 2. The intercom shall be a small lightweight battery powered unit suitable for attachment to components of the aircrew ensemble without modification of the garments. The intercom shall be suitable for use both indoors and outdoors and while the user is donning and doffing his chemical, biological, radiological (CBR) protective ensemble. The intercom shall provide the user with good auditory response to his immediate surroundings while leaving both hands free. Battery ON/OFF switching shall be activated by the insertion of the headset jack plug into either of the jack sockets. Battery replacement shall be readily effected with CBR gloved hands. The design shall permit decontamination using materials and procedures of Army FM 3-5 without damage to or deterioration of the intercom. The design shall include provisions for preventing moisture and contaminants from entering and collecting inside the intercom and to facilitate inspection, servicing, maintenance and cleaning.

3.5.1 Storage life. The intercom, less battery, shall be designed to meet the performance requirements of this specification after five years storage under worldwide climatic conditions specified in MIL-STD-210, when packaged as specified in MIL-STD-794.

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3.5.2 Interchangeability. All parts, subassemblies and assemblies having the same part number shall be identical as specified in DOD-STD-480 and DOD-STD-100. When any unit or part is interchanged, the intercom shall meet all performance requirements without adjustment or modification of any part or subassembly.

3.5.3 Printed wiring boards Printed wiring boards shall be designed as specified in MIL-P-55110.

3.5.4 Reverse polarity. The intercom shall not be damaged by the inadvertant reversal of the battery polarity.

3.5 5 Sidetone. The amplifier shall provide sidetone into the aircrew helmet headset specified in NAVAIR 13-1-6.7.

3.6 Construction.

3.6.1 Standard parts. Military standard parts shall be used whenever they are suitable for the purpose. Standard parts shall be as specified in MIL-E-5400.

3.6.1.1 Nonstandard parts approval. Request for approval for the use of nonstandard parts shall be as specified in MIL-STD-965, Procedure I.

3.6.2 Selection of printed wiring boards. The selection of parts and material shall be as specified in MIL-STD-454, Requirement 17 or with the approval of the acquiring activity Microelectric technology should be utilized to the fullest extent possible Components shall be rated so that the intercom shall function within the limits of this specification after extended inoperative periods while carried in military aircraft and subjected to the varied range of environmental conditions defined herein. Nonrepairable subassemblies, as outlined in MIL-E-5400, shall be used when practical.

3.6.2.1 Printed wiring board modifications. Printed wiring board modifications shall be performed as specified in MIL-P-28809

3.6.2.2 Electronic components. All electronic compnents shall be as specified in MIL-STD-454.

3.6.3 Insulating material. Electrical insulating materials shall be as specified in MIL-STD-454, Requirement II Insulation for power circuits shall be arc resistant as specified in MIL-STD-454, Requirement 26.

3.6.4 Encapsulation and embedment. Materials for encapsulation and embedment shall be as specified in MIL-STD-454, Requirement 47.

3.6.5 Semiconductor devices. Semiconductor devices shall be selected and applied as specified in MIL-STD-454, Requirement 30. Microelectronic devices shall be selected and applied as specified in MIL-STD-454, Requirement 64. Use of nonstandard parts shall be substantiated by reliability data or testing to assure adequate device reliability for the application.

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3.6.6 Resistors. Resistors shall be as specified in MIL-STD-454, Requirement 33.

3.6.7 Capacitors. Capacitors shall be chosen and applied as specified in MIL-STD-454, Requirement 2.

3.6.8 Soldering. Soldering shall be as specified in MIL-STD-454, Requirement 5.

3.6.9 Relays. Relays shall be as specified in MIL-STD-454, Requirement 57.

3.6.10 Wiring. Wiring shall be as specified in MIL-W-16878, 20 gauge wire, type B, or equal.

3 7 Hardware.

3.7.1 Fasteners. All fasteners, including bolts, screws, nuts, rivets, washers and hinges shall be as specified in SD-24.

3 8 Quantitative reliability requirements. When tested in accordance with MIL-STD-781 as specified in 4.8.2.10, the mean time between failure (MTBF) for the intercom, based on an exponential distribution of failure times, the lower single sided 90 percent confidence limit, shall be 500 hours (excluding batteries) The performance requirements specified in 3.9.1, 3.9 2, 3.9.3, 3.9.4, 3.9.5, 3.9.6, 3.9.7, 3.9.8, and 3.9 9 shall be the reliability requirements for determining mean time between failures.

3.9 Performance.

3.9.1 ON/OFF switching. The intercom shall be tested as specified in 4.8.2 1 Battery ON/OFF switching shall be achieved by the insertion of the headset assembly jack plug into either of the microphone/telephone sockets.

3.9.1 1 Endurance The intercom switch shall be cycled 1000 times and shall operate with no failure when tested as specified in 4.8.2.1.1.

3.9.2 Output power and current consumption. The intercom shall be tested as specified in 4 8 2.2 Current consumption shall be not greater than 16 mA. d.c in receive mode and 50 mA. d c. in transmit mode

3.9.3 Frequency response. The intercom shall be tested as specified in 4.8.2.3. The audio frequency response of the amplifier shall be better than ± 6 dB. relative to 2000 Hz.

3 9.4 Distortion The intercom shall be tested as specified in 4.8.2.4. The distortion shall be less than 5 percent

3 9.5 Peak output level. The intercom shall be tested as specified in 4.8.2.5 The amplifier shall produce a minimum of 0.4 volts rms before clipping occurs.

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3.9 6 Electrical input to acoustical output. The intercom shall be tested as specified in 4.8.2.6. The mean sound pressure level at 24 in. from the speaker unit shall be not less than 80 dB. above 20 uPa.

3.9.7 Acoustical input to electrical output. The intercom shall be tested as specified in 4.8.2.7. The mean electrical output shall be not less than 1.0 mW.

3.9.8 Frequency response - Transducer as loudspeaker. The electrical input to acoustic output frequency response of the transducer and amplifier combined shall be better than ± 10 dB. The output at 300 Hz. shall be greater than 14 dB. below that at 1500 Hz. when tested as specified in 4.8.2.8

3.9.9 Frequency response - Transducer as microphone. The acoustic input to electrical output frequency response of the transducer and amplifier combined shall be better than ± 10 dB. The output at 300 Hz. shall be greater than 14 dB below that at 1500 Hz. when tested as specified in 4.8.2.9.

3 9 10 Environmental. Prior to and at the completion of each environmental test, the intercom shall be tested and meet the requirements of 3.9.1, 3.9.2, 3.9 3, 3 9.4, 3.9.5, 3.9.6, 3.9.7, 3.9.8, and 3.9.9

3 9.10 1 High temperature. When tested as specified in 4 8.3.1, the intercom shall operate without failure

3 9.10 2 Low temperature When tested as specified in 4.8.3.2, the intercom shall operate without failure.

3 9.10.3 Salt fog When tested as specified in 4.8 3.3, the intercom shall operate without failure and show no structural degradation.

3 9.10 4 Rain When tested as specified in 4 8 3 4, the intercom shall operate without failure.

3.9.10.5 Fungus When tested as specified in 4.8 3 5, the intercom shall not support fungal growth and shall operate without failure.

3.9.10.6 Low pressure and high humidity. When tested as specified in 4.8.3.6, the intercom shall operate without failure.

3.9.10.7 Dust When tested as specified in 4.8.3.7 the intercom, shall operate without failure

3 9.10.8 Normal acceleration. When tested as specified in 4.8.3.8, the intercom shall operate without failure.

3 9.10 9 High temperature, diurnal cycle and high humidity. When tested as specified in 4.8.3.9, the intercom shall operate without failure.

3 9 10 10 Shock When tested as specified in 4.8.3.10, the intercom shall operate without failure, and show no structural degradation.

3 10 Components The intercom shall consist of the components specified in 3.10.1 and 3.10.2 and other parts not specifically mentioned which are essential to form a complete, assembled, and operating intercom.

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3.10.1 Case. The case shall be smooth contoured to prevent both the accumulation of liquid contaminants and abrasion or damage to other components of the aircrew ensemble. Corners and sharp projections shall be avoided.

3.10.2 Battery housing The battery housing shall be isolated and sealed from the environment and the other internal components of the intercom to prevent damage from battery leakage or venting, as shown on Figure 2.

3.10.3 Snap hook. The snap hook attaches to the rear of the case to allow connection to the SV-2 survival vest, as shown on Figure 2.

3.10.4 Electrical component housing cover. A cover shall be capable of being removed from the case. Removal of the cover shall give access to the interior of the unit and all electrical components except the battery.

3 10 5 Switch. A switch shall be accessible and operable by the wearer of the PASS utilizing a CBR gloved hand. The switch shall enable the wearer to select the following modes of communications.

a Communication between the 2 headset positions with the transducer switched off.
b Communication between the 2 headset positions together with the reception of signals from the transducer (operating in the microphone mode) at both headset positions.
c Speech through the transducer (operating in the loudspeaker mode) from both headset positions

3.10.6 Battery A battery shall be used to supply electrical power for the intercom. The operational life of the battery shall be not less than six hours of continuous operation with a transducer microphone/loudspeaker ratio of 9:1

3.10.7 Sockets. Two microphone/telephone sockets, as specified in MIL-C-9177/1, shall be located on the intercom, and shall allow for the insertion of two communications cords, as shown on Figure 2.

3.11 Dimensions. The dimensions of the intercom shall be as shown on Figure 2

3.12 Weight The weight of the intercom shall be not greater than 12 ounces.

3.13 Color. The intercom shall be colored black, approximately matching color number 37038 as specified in FED-STD-595.

3.14 Nameplates and identification markings. Nameplate approval, equipment identification and serial number assignment marking shall be as specified in MIL-N-18307.

3.15 Configuration. The intercom furnished shall be produced under a program for managing system configuration as specified in DOD-STD-480, MIL-STD-483 and MIL-STD-490

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3.16 Workmanship. The intercom shall be manufactured in a manner that results in a product of uniform excellent quality free from defects which could effect safety, performance, reliability or durability. The intercom shall be free from contamination and imperfections such as dirt, grease, oil, foreign matter, malfunctioning of hardware and damage such as cracks, discoloration, stress flaws and sharp edges.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4 1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absense of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.2 Classification of inspection. The inspection requirements specified herein are classified as follows:

a. Qualification inspection (see 4.3)
b. First article inspection (see 4.4).
c. Quality conformance inspection (see 4.5)
d. Quality conformance verification inspection (see 4.6).

4.3 Qualification inspection Qualification inspection shall be performed in the order specified in Table I, consisting of all the tests and examinations listed in this specification

4.3.1 Qualification samples. Qualification samples shall consist of fifteen intercoms. Samples shall be forwarded to a test facility set forth in the letter of authorization to submit samples (see 6.3). The samples shall be plainly identified by securely attached durable tags marked with the following information:

Samples submitted by (name) (date) for qualification inspection as specified in MIL-I-85832(AS) and number under authorization (reference authorizing letter and number) (see 6.3)

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4.3.1.1 Disposition of samples. Upon completion of qualification inspection, all samples shall be consumed or destroyed and shall not be considered as part of the quantity to be delivered under contract.

4.3.2 Retention of qualification. To retain qualification, the contractor shall forward a report at six month intervals to the qualifying activity. The qualifying activity shall establish the initial reporting date. The report shall consist of a summary of the results of the tests performed, indicating the number of samples that have passed and the number that have failed during the six month period. If the summary of the test results indicates non conformance with specification requirements, and corrective action acceptable to the qualifying activity has not been taken, action may be taken to remove the failing product from the qualified products list. Failure to submit the report within 30 days after the end of each six month period may result in loss of qualification for the product. In addition to the periodic reports, the contractor shall immediately inform the qualifying activity of inspection data indicating failure of the qualified product to meet this specification. In the event that no production occurred during the reporting period, a report shall be submitted certifying that the company still has the capabilities and facilities necessary to produce the item. If during two consecutive reporting periods, there has been no production, the manufacturer may be required at the discretion of the qualifying activity, to submit the qualified products to testing in accordance with qualification inspection requirement and the reason for no production.

4.4 First article inspection. First article inspection shall be performed in the order specified in Table II.

4 4 1 First article samples. After award of the contract or order, the manufacturer shall submit 3 intercoms The samples shall be representative of the construction, workmanship and materials to be used during production. When a manufacturer is in continuous production of these couplings from contract to contract, submission of further first article samples on a new contract may be waived at the discretion of the acquiring activity (see 6.2). Approval of the first article samples or the waiving of the first article inspection does not preclude the requirements of submitting to the quality conformance inspection. The first article inspection samples shall be furnished to the Government as directed by the contracting officer (see 6 2)

4.4.1 1 Disposition of samples. Upon completion of the first article inspection, one approved intercom shall be returned to the manufacturer for use in monitoring production. The other samples shall be consumed or destroyed in the first article inspection and shall not be considered as part of the quantity to be delivered under contract.

4.5 Quality conformance inspection. Quality conformance inspection shall consist of all the examinations and tests specified in Table II. The sampling and inspection levels shall be as specified in MIL-STD-105.

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4.5.1 Inspection lot. An inspection lot size shall be expressed in units of one intercom made by one manufacturer under essentially the same conditions and from the same materials and components.

4.5.1.2 Packaging. An inspection lot size shall be expressed in units of one fully prepared shipping container, containing intercoms, fully prepared for delivery, made from essentially the same materials and components. The sample unit shall be one shipping container, containing intercoms, fully prepared for delivery, with the exception that the shipping container need not be sealed.

4.5.2 Sampling for tests and examination of intercoms. The sample size, acceptance criteria, tests and examinations required for the intercoms shall be as specified in Table III.

4.6 Quality conformance verification inspection. At the option of the Government and upon completion of the quality conformance inspection requirements of 4.5, a random sample, one for every 150 or fraction thereof, shall be selected from each lot of intercoms (see 6.2.1). Each intercom, selected as a sample unit, shall be forwarded to a laboratory designated at the time of the award. The designated laboratory shall conduct any or all the tests and examinations of 4.8 of this specification. The serial numbers of the units in the lot, represented by the sample units, shall be included with the data accompanying the samples to the laboratory. The Government activity responsible for conducting the quality conformance verification program shall report the results of the tests and examinations to the designated inspection and acceptance activity specified in the acquisition document. Final acceptance of the lot from which the sample units were selected shall be based upon successful completion of the inspection program by the cognizant Quality Assurance Representative/ Specialist at the contractor's facility; applying the applicable acceptance criteria specified in Table III.

4.6.1 Rejected lots. If an inspection lot is rejected, the manufacturer may rework it to correct the defects, or screen out the defective units, and resubmit for reinspection. Resubmitted lots shall be inspected using tightened inspection and shall not thereafter be tendered for acceptance unless the former rejection or requirement of correction is disclosed. Such lots shall be seperate from new lots and shall be clearly identified as reinspected lots.

4.7 Test conditions.

4.7.1 Temperature and pressure. Unless otherwise specified in the contract, inspections shall be conducted at local ambient temperature and barometric pressure. Corrections shall be made to provide agreement with the temperature and pressure calibration of the instruments. Inspection data provided by an instrument not calibrated to normal temperature and pressure (NTP) conditions shall be corrected to determine NTP conditions NTP conditions are 70°F and 29.92 in. Hg.

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4.8 Inspection methods.

4.8.1 Visual examination.

4.8.1.1 Intercom. Every intercom, selected as a sample unit from the lot, shall be visually examined for major and minor defects to determine conformance to this specification. The classification of defects (see Table IV), shall be used to classify the defects found.

4.8.1.1.1 Dimensions. Each intercom, selected as a sample unit from the lot, shall be checked dimensionally to determine conformance to the dimensions herein.

4.8.1.2 Packaging. Each of the fully prepared shipping containers, containing intercoms, selected as a sample unit from the lot, shall be examined to determine that the packaging, packing and marking conform to this specification The list of defects (see Table V), shall be used to enumerate the defects found.

4 8 2 Performance.

4.8.2.1 ON/OFF switching. A headset assembly jack plug shall be plugged into one of the jack sockets and then into the other jack socket and checked for function The intercom shall meet the requirements specified in 3.9.1.

4.8.2.1.1 Endurance. The intercom switch shall be cycled 1000 times. A cycle shall consist of selecting the three modes of communication as specified in 3.10.5. At the conclusion of the 1000 cycles, the switch shall meet the requirements specified in 3.9.1.1.

4.8.2 2 Output power and current consumption. The audio power into 10 ohms ± 2% resistive load connected across the telephone terminals of each jack socket shall be 5.0 mW. ± 2 dB. with an input voltage of 250 uVrms. into the terminals of either jack socket. The intercom shall meet the requirements specified in 3 9 2.

4.8.2.3 Frequency response. With an input voltage of 250 mVrms. into the terminals of either jack socket, the audio frequency response of the amplifier shall be tested over the frequency range 300 Hz. to 3500 Hz The intercom shall meet the requirements specified in 3.9.3.

4.8.2.4 Distortion. With an input voltage of 250 uVrms into the terminals of either jack socket, the outputs shall be loaded with 10 ohm telephone and 12 ohm transducer. The intercom shall meet the requirements specified in 3.9.4.

4.8.2.5 Peak output level. The amplifier shall produce not less than 0.4 Vrms into a 10 ohm ± 2 percent resistive load before clipping occurs. The intercom shall meet the requirements specified in 3.9.5.

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4.8.2.6 Electrical input to acoustical output. With an input voltage of 250 uVrms. into the terminals of either jack socket, over the frequency band 1500 Hz. to 3000 Hz. the accoustical output shall be measured. The intercom shall meet the requirements specified in 3.9.6.

4.8.2.7 Acoustical input to electrical output. A sound pressure level of 80 dB above 20 uPa shall be applied at the face of the transducer. The mean electrical output over the frequency band 1500 Hz. to 3000 Hz. into 10 ohm ± 2% resistive load shall meet the requirements specified in 3.9.7.

4.8.2.8 Frequency response - Transducer as loudspeaker. With an input voltage of 250 uVrms into the terminals of either jack socket, the electrical input to acoustic output frequency response of the transducer and amplifier combined shall be measured between 1500 Hz. to 3500 Hz. The intercom shall meet the requirements specified in 3.9.8.

4.8.2.9 Frequency response - Transducer as microphone. The transducer shall be sited in a field of 80 dB. sound pressure level above 20 uPa. The intercom shall meet the requirements specified in 3.9.9.

4.8.2.10 Reliability. Five intercom sets shall be tested using Test Plan IVC of MIL-STD-781. (These units shall not be the same units used for environmental testing.) Each intercom shall be subjected to 315 hours of testing. The test cycle shall be 14 hours in duration and shall consist of a 7 hour hot mission and a 7 hour cold mission. Cold mission temperature levels shall start at +131°F, rise to +158°F for 1 hour followed by 1 hour at +131°F with the remaining 5 hours atr -14°F. Hot mission temperature levels shall start at -14°F lowered to -40°F for 1 hour, followed by 1 hour at -14°F, with the remaining 5 hours at +131°F The intercoms shall be powered by a remote power supply and shall be subjected to operational tasks as specified in 4.8.2.1, 4.8.2.2., 4.8.2.3, 4.8.2 4, 4.8.2.5, 4.8.2.6, 4.8.2.7, 4.8.2.8, and 4.8.2.9 and meets the requirements of 3.8.

4.8.3 Environmental tests Environmental tests shall be conducted in the sequence specified in Table VI. Prior to and at the completion of each environmental test the intercom shall be tested as specified in 4.8.2 1, 4 8.2.3, 4.8 2.4, 4 8 2.5, 4 8 2.6, 4 8.2.7, 4.8.2.8, and 4.8.2.9 and shall meet the requirements of 3.9.10.

4.8.3.1 High temperature. The intercom shall be tested as specified in MIL-STD-810, Method 501.2, Procedure II, Operation. The intercom shall be placed in a climatic chamber. The chamber temperature shall be raised to 158°F at a rate of 5F° per minute. The temperature shall be maintained for a period of one hour. The relative humidity shall be not greater than 75 percent. The temperature shall then be brought to 131°F at a rate of 5F° per minute and maintained for a period of six hours. The intercom shall meet the requirements specified in 3 9.10 and 3.9.10.1.

4.8.3.2 Low temperature. The intercom shall be tested as specified in MIL-STD-810, Method 502.2, Procedure II, Operation. The intercom shall be placed in a climatic chamber. The chamber temperature shall be lowered to -40°F at a rate of 5°F per minute The temperature shall be maintained for a period of one hour. The temperature shall then be brought to -14°F at a rate of 5°F per minute and maintained for a period of six hours. The intercom shall meet the requirements specified in 3.9.10 and 3.9.10.2.

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4.8.3.3 Salt fog. The intercom shall be tested as specified in MIL-STD-810, Method 509.2. The test shall consist of four alternating periods of 24 hours exposure and 24 hours drying time. The intercom shall meet the requirements specified in 3.9.10 and 3.9.10.3.

4.8.3.4 Rain. The intercom shall be tested as specified in MIL-STD-810, Method 506.2, Procedure I, Blowing Rain, for a period of one hour at a rate of four inches per hour. The intercom shall meet the requirements specified in 3.9.10 and 3.9.10.4.

4.8.3 5 Fungus. The intercom shall be tested as specified in MIL-STD-810, Method 508.3 for a period of 28 days. The intercom shall meet the requirements specified in 3.9.10 and 3.9.10.5.

4.8.3.6 Low pressure and high humidity. The intercom shall be tested as specified in MIL-STD-810, Method 500.2, Procedure II, Operation. The intercom shall be placed in a climatic chamber. The chamber temperature shall be reduced to -14°F at a rate not exceeding 5F° per minute and the condition maintained for a period of 30 minutes. The pressure in the chamber shall then be reduced to .7 standard atmospheres over a period of 15 minutes and maintained at this pressure for 30 minutes The chamber humidity shall then be raised and maintained at, or close to saturation. The chamber temperature shall then be raised to 32°F and the pressure increased to NTP pressure at an approximately linear rate over a period of 15 minutes. The chamber temperature shall then be raised at an approximately linear rate to 86°F with the humidity maintained at or near saturation, over a period of 30 minutes. The chamber conditions shall be maintained for a period of not less than one hour and until all frost has melted. The intercom shall meet the requirements specified in 3.9.10 and 3.9.10.6.

4 8.3 7 Dust. The intercom shall be tested as specified in MIL-STD-810, Method 510.2, Procedure I, Blowing Dust for a period of six hours at 73°F and 300 ft./min. air velocity and six hours at 131°F and 1750 ft./min. air velocity. Relative humidity shall be not greater than 30 percent. The intercom shall meet the requirements specified in 3 9 10 and 3.9.10.7.

4.8.3 8 Normal acceleration. The intercom shall be tested as specified in MIL-STD-810, Method 513.3, Procedure II, Operational Test. The acceleration shall be six g's in each of the six directions. The intercom shall meet the requirements specified in 3.9.10 and 3 9.10.8.

4.8.3.9 High temperature, diurnal cycle and high humidity. The intercom shall be tested as specified in MIL-STD-810, Method 507.2, Procedure II, Cycle 4, normal test duration. The intercom shall be examined every five days and shall meet the requirements specified in 3.9.10 and 3.9.10.9.

4.8.3 10 Shock The intercom shall be tested as specified in MIL-STD-810, Method 516.3, Procedure IV, Transit Drop The intercom shall be dropped on each of its six major faces with its lower surface 48 inches above a flat concrete surface The intercom shall meet the requirements specified in 3.9.10 and 3.9.10.10.

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5. PACKAGING

5.1 Preservation. Preservation shall be level A or Commercial (see 6.2.1).

5.1.1 Level A. The intercom shall be preserved Method IIC as specified in MIL-P-116.

5 1.1.1 Cleaning and drying Cleaning shall be "C-1", any applicable process, as specified in MIL-P-116 and any drying process of MIL-P-116 may be applied.

5.1.1.2 Preservative. No preservative is required.

5.1.1.3 Unit packaging. The intercom shall be cushioned with material as specified in PPP-C-795, or PPP-C-1842 and shall be placed with an appropriate amount of desiccant into a sealed bag conforming to MIL-B-117, Types I, II or III, Classes E, F or G, Styles 1, 2 or 3 The bag shall then be placed in a PPP-B-636, Type CF, Weather Resistant Unit Container.

5.1.2 Commercial. The intercom shall be preserved and unit packed in accordance with ASTM D 3951.

5.2 Packing Packing shall be Level A, B or Commercial (see 6 2 1)

5.2.1. Level A. The intercom, preserved as specified in 5.1.1.3, shall be packed in containers conforming to Table I of MIL-STD-794, as specified for Level A.

5 2 2 Level B The intercom, preserved as specified in 5.1.1.3, shall be packed in containers conforming to Table I of MIL-STD-794, as specified for Level B.

5.2.3 Commercial. The intercom preserved and unit packed as specified above shall be provided packing in accordance with ASTM D 3951

5.3 Marking All unit packages and shipping containers shall be marked as specified in MIL-STD-129, including bar code markings as specified in MIL-STD-129 and MIL-STD-1189.

6 NOTES

6.1 Intended use. The intercom covered by this specification is intended to provide the wearer of the PASS in CBR contamination environments with the ability to converse:

a. with another simlarly dressed person.
b. with the wearer of a respirator with no microphone/telephone equipment.
c. with an unprotected person.

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6.2 Ordering data.

6.2.1 Acquisition requirements. Acquisition documents should specify the following:

a Title, number and date of this specification.
b. Applicable Government part number.
c Whether first article inspection is waived or the name and address of the first article inspection laboratory and the name of the Government activity responsible for conducting the first article inspection program (see 4.4.1).
d If quality conformance verification inspection is required, the name and address of the designated laboratory and the Government activity responsible for the verification program (see 4.6).
e. Applicable levels of preservation, packaging and packing (see 5.1 and 5.2).

6.3 Qualification. With respect to products requiring qualification, awards will be made only for products which are, at the time set for opening of bids, qualified for inclusion in the applicable Qualified Products List whether or not such products have actually been so listed by that date. The attention of contractors is called to these requirements. Manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification in order that they might be eligible to be awarded contracts or purchase orders for the products covered by this specification. The activity responsible for the Qualified Products List is the Commander, Naval Air Systems Command (Code 51122), Department of the Navy, Washington, DC 20361-5110; however, authorization for qualification of products shall be obtained from the Commander, Naval Air Development Center, Warminster, PA 18974, (Code 6031). Prior to submission of the samples for qualification inspection, the manufacturer shall submit a request to the Naval Air Development Center (Code 6031) indicating a date on which the samples can be forwarded and also request an authorization number to accompany the samples.

6.4 First article. When a first article inspection is required, the item should be a first article sample. The first article should consist of 3 complete intercoms The contracting officer should include specific instructions in acquisition documents, regarding arrangements for examinations, tests and approval of the first article. Invitations for bids should provide that the Government reserves the right to waive the requirements for samples for first article inspection to those bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract.

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6.5 Key word listing.

Acoustical
Audio frequency amplifier
Biological
CBR protective
Chemical
Frequency response
Helicopter aircrewman
Intercom
Jack socket
Microswitch
Radiological
Transducer

Preparing activity:
Navy - AS
(Project 5965-N198)

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Table I. Qualification inspection.

Inspection	Requirement	Test method	Sample number 1 2 3
Visual	-	4.8.1	X X X
ON/OFF switching and reverse polarity	3.9.1	4.8.2.1	X X X
Endurance	3.9.1.1	4.8.2.1.1	X X X
Output power and current consumption	3.9.2	4.8.2.2	X X X
Frequency response	3.9.3	4.8.2.3	X X X
Distortion	3.9.4	4.8.2.4	X X X
Peak output level	3.9.5	4.8.2.5	X X X
Electrical input to acoustical output	3.9.6	4.8 2.6	X X X
Acoustical input to electrical output	3.9.7	4.8.2.7	X X X
Frequency response/transducer as loudspeaker	3.9.8	4.8.2.8	X X X
Frequency response/transducer as microphone	3.9.9	4.8 2.9	X X X
High temperature	3.9.10 1	4.8.3.1	See Table VI
Low temperature	3.9.10.2	4.8.3.2	See Table VI
Salt fog	3.9.10.3	4.8.3.3	See Table VI
Rain	3.9.10.4	4.8.3.4	See Table VI
Fungus	3.9.10.5	4.8.3.5	See Table VI
Low pressure and high humidity	3.9.10.6	4.8.3.6	See Table VI
Dust	3.9.10.7	4.8.3.7	See Table VI
Normal acceleration	3.9.10 8	4.8.3.8	See Table VI
High temperature, diurnal cycle and high humidity	3.9.10.9	4.8.3.9	See Table VI
Shock	3.9.10.10	4.8.3.10	See Table VI

Table II First article inspection.

Inspection	Requirement	Test method	Sample number 1 2 3
Visual	-	4.8.1	X X X
ON/OFF switching and reverse polarity	3.9.1	4.8.2.1	X X X
Endurance	3.9.1.1	4.8.2.1.1	X X X
Output power and current consumption	3.9.2	4.8.2.2	X X X
Frequency response	3.9.3	4 8.2.3	X X X
Distortion	3.9.4	4.8.2.4	X X X
Peak output level	3.9.5	4.8.2.5	X X X
Electrical input to acoustical output	3.9.6	4 8.2 6	X X X
Acoustical input to electrical output	3.9.7	4.8.2 7	X X X
Frequency response/transducer as loudspeaker	3.9.8	4.8.2.8	X X X
Frequency response/transducer as microphone	3.9 9	4.8.2 9	X X X

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Table III. Sample size, acceptance criteria, tests and examination of the intercom.

Inspection	Method	Sample size	Acceptance criteria
Visual	4.8.1	Normal level II S	AQL 2.5% major AQL 4.0% minor
ON/OFF switching and reverse polarity	4.8.2.1	Normal level II S	AQL 1.0%
Output power and current consunption	4.8.2.2	Normal level II S	AQL 1.0%
Frequency response	4.8.2.3	Normal level II S	AQL 1.0%
Distortion	4.8.2.4	Normal level II S	AQL 1.0%
Peak output level	4.8.2.5	Normal level II S	AQL 1.0%
Electrical input to acoustical output	4.8.2.6	Normal level II S	AQL 1.0%
Acoustical input to electrical output	4.8.2.7	Normal level II S	AQL 1.0%
Frequency response - transducer as loudspeaker	4.8.2.8	Normal level II S	AQL 1.0%
Frequency response - transducer as microphone	4.8.2.9	Normal level II S	AQL 1.0%
Packaging	4.8.1.2	Normal level II S	AQL 2.5%

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Table IV. Classification of defects for visual examination of the intercom.

Major	Minor
101. Material imperfections-foreign matter embedded.	201. Marking missing, insufficient, incorrect, illegible or not permanent.
102. Surface unclean, rough, misaligned, or containing cracks, nicks or other flaws.	
103. Any component missing, malformed, fractured, or otherwise damaged.	
104. Any component loose or otherwise not securly retained.	
105. Incorrect assembling or improper positioning of components.	
106. Any functioning part that works with difficulty.	
107 Faulty workmanship or other irregularities	

Table V. List of defects for packaging.

Item	Defects
Exterior and interior markings.	Missing, incorrect, incomplete, illegible; of improper size, location, sequence; or method of application; markings not the same on the interior and exterior containers.
Packaging and packing materials	Any non-conforming component; any component missing, damaged, or otherwise defective.
Workmanship.	Inadequate application of the components such as incomplete closure of the unit package, intermediate package. container flaps, loose strapping, etc., bulging or distortion of the containers.
Exterior and interior weight or content.	Number per container is more or less than required; gross or net weight exceeds the requirements.

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Table VI. Environmental qualification test sequence (intercom).

Test	Unit Number *						
	1	2	3	4	5	6	7
Visual	X	X	X	X	X	X	X
Low Pressure/High Humidity	X	X					
Low Temperature	X	X					
High Temperature	X	X					
Blowing Dust	X	X					
Rain			X	X			
Fungus			X	X			
Salt Fog			X	X			
Acceleration					X	X	
High Temp/Diurnal/ High Humidity					X	X	
Shock - Transit Drop							X

*These units shall not include the five intercoms used in reliability testing (4.8.2.10).

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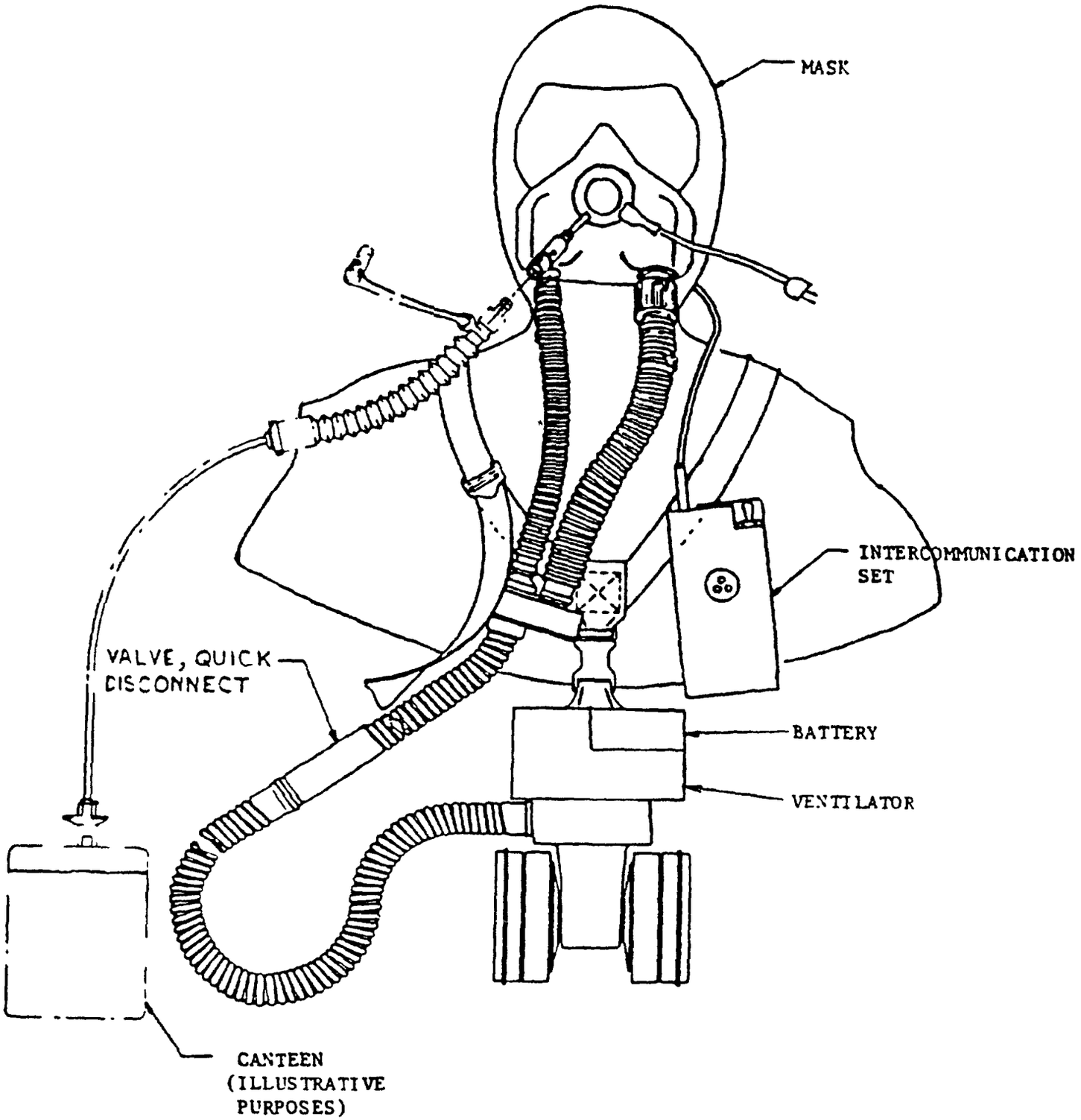


FIGURE 1. PROTECTIVE ASSEMBLY, CBR, HELICOPTER AIRCREWMEMBER

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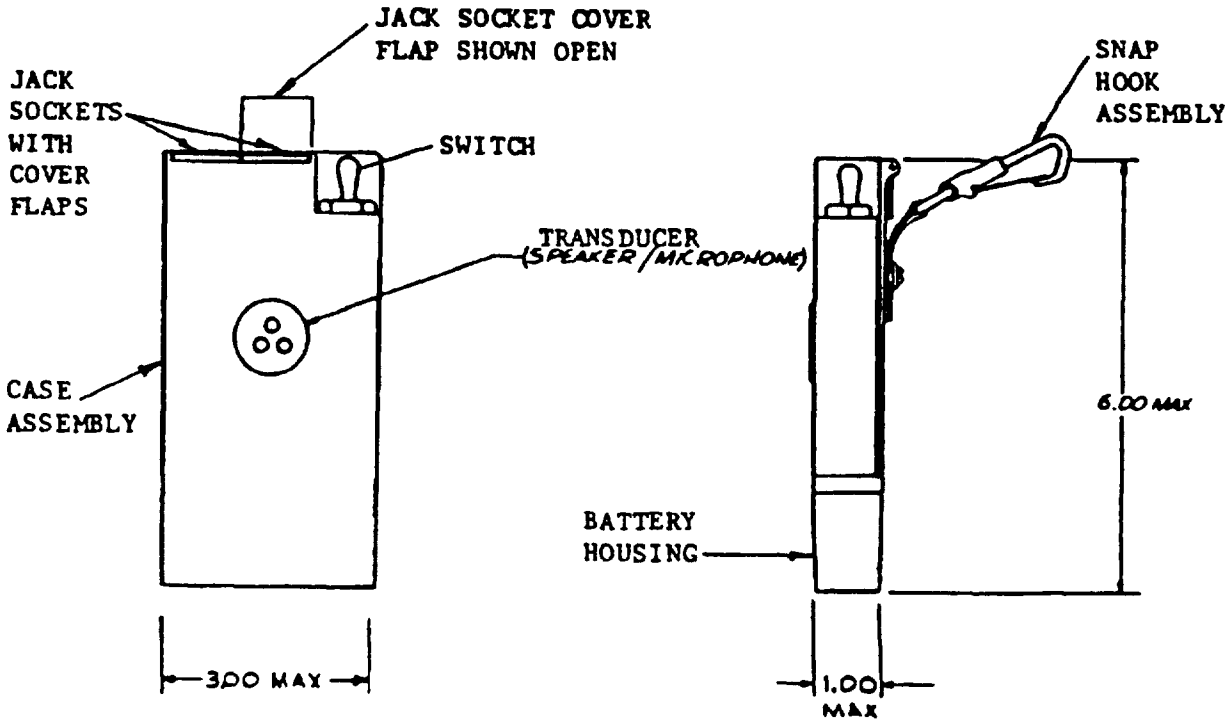


FIGURE 2. INTERCOMMUNICATION SET

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c. Reason/Rationale for Recommendation

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